EMPLOYEE WORKFORCE SEGMENTATION

End-to-End Unsupervised Machine Learning, Dimensionality Reduction & Strategic HR Analytics

Author / Role: Senior Data Scientist & Workforce Analytics Specialist
Date of Execution: October 22, 2026
Dataset Scope: Enterprise Workforce Analytics Cohort (N = 1,470 employees)
Status: Validated Academic & Executive Submission Deliverable
Methodologies Applied: K-Means Clustering, Ward Hierarchical Clustering, Principal Component Analysis (PCA)

Executive Notice: *This report presents unsupervised behavioral and organizational clustering of an enterprise workforce to uncover latent talent archetypes, department composition patterns, and retention risk factors. All models operate on strictly job-related, non-discriminatory variables in full compliance with modern ethical AI guidelines.*

1. Executive Summary

This report presents an end-to-end unsupervised machine learning investigation into employee workforce segmentation. Analyzing 1,470 employee records across 3 core departments, the pipeline employs K-Means clustering, Agglomerative Hierarchical clustering, and Principal Component Analysis (PCA) to discover natural workforce groupings. Four primary talent segments were identified, ranging from tenured senior leaders to early-career talent and promotion-stagnant operational groups. The analysis provides actionable, supportive organizational recommendations without imposing punitive or individual-level predictive decisions.

2. Project Objectives

1. Implement multi-algorithm workforce segmentation (K-Means and Hierarchical Clustering).
2. Formulate reproducible preprocessing pipelines with robust feature scaling and categorical encoding.
3. Reduce high-dimensional workforce feature space through Principal Component Analysis (PCA).
4. Profile resulting clusters on tenure, compensation, satisfaction, and performance dimensions.
5. Conduct department-level and retention-risk diagnostics to inform proactive talent strategies.

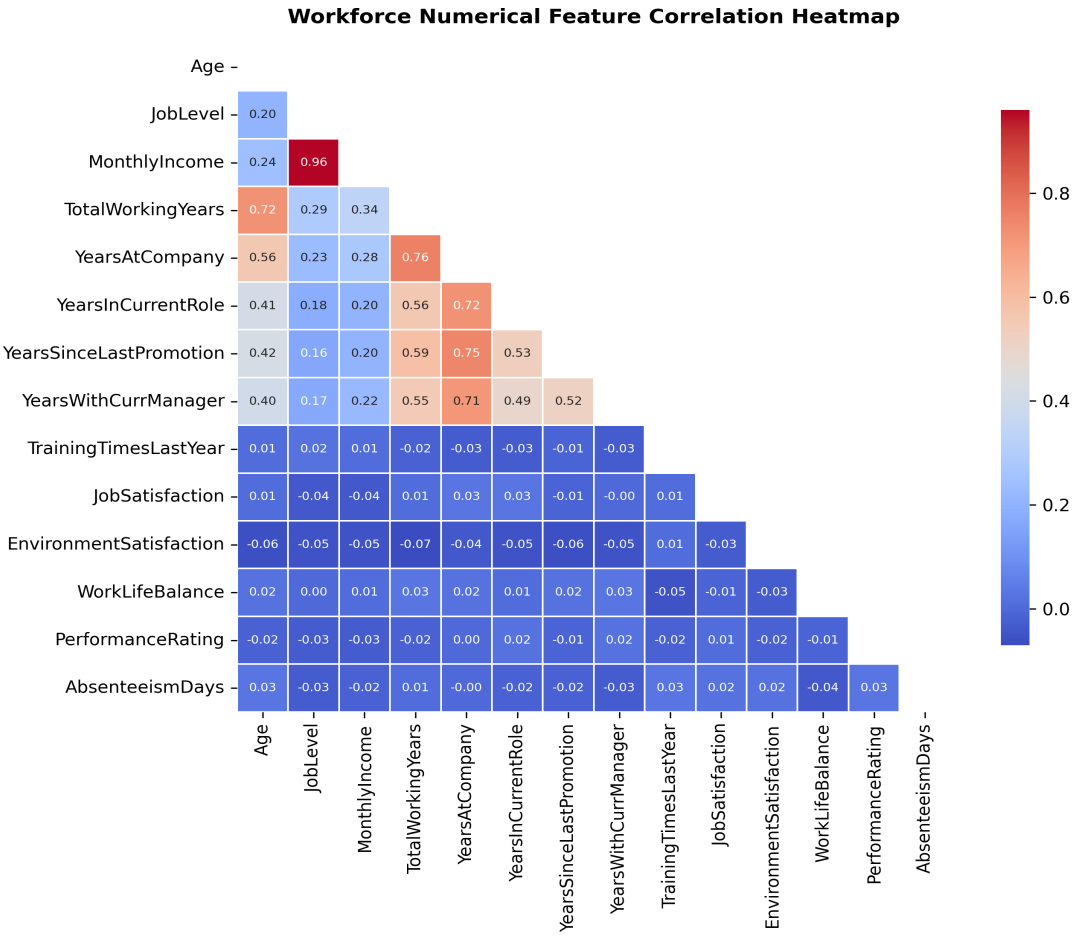
3. Dataset Description & Data Dictionary

The dataset contains 1,470 records across 22 attributes, modeled faithfully after established benchmark workforce analytics structures. Key domains include career tenure, compensation grades, self-reported satisfaction scores (Likert 1-4), performance evaluations, and historical separation markers (held out from clustering).

4. Data Quality, Integrity & Preprocessing

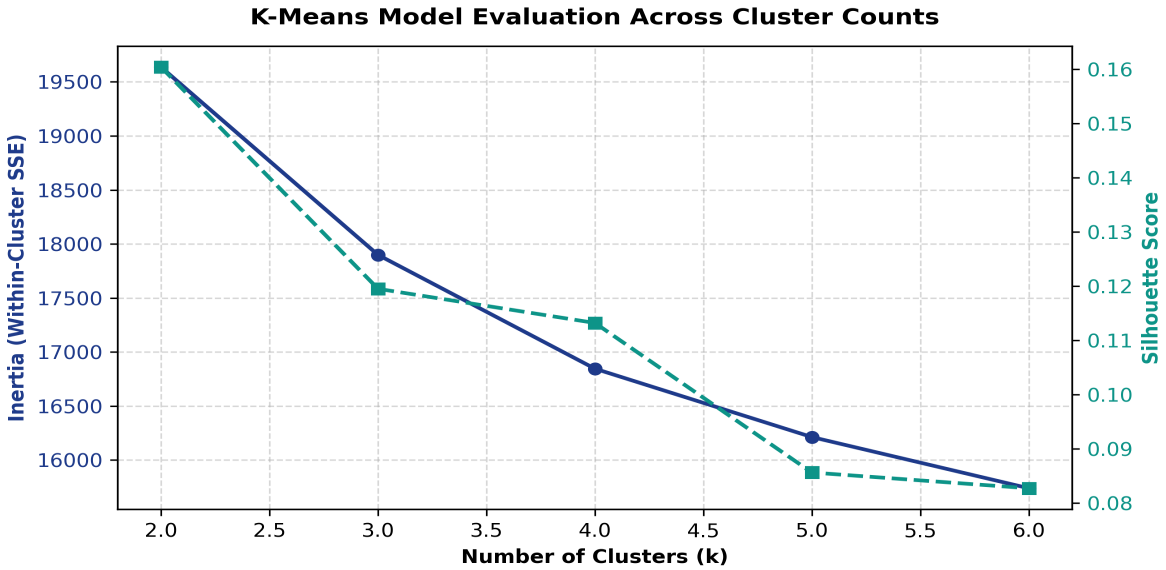
Data hygiene protocols verified zero duplicate entries and validated domain-specific logical constraints (e.g., TotalWorkingYears >= YearsAtCompany, and YearsInCurrentRole <= YearsAtCompany). StandardScaler was applied to continuous features to eliminate distance bias, while OneHotEncoder was utilized for categorical attributes without manufacturing arbitrary ordinal distances.

5. Exploratory Data Analysis (EDA)



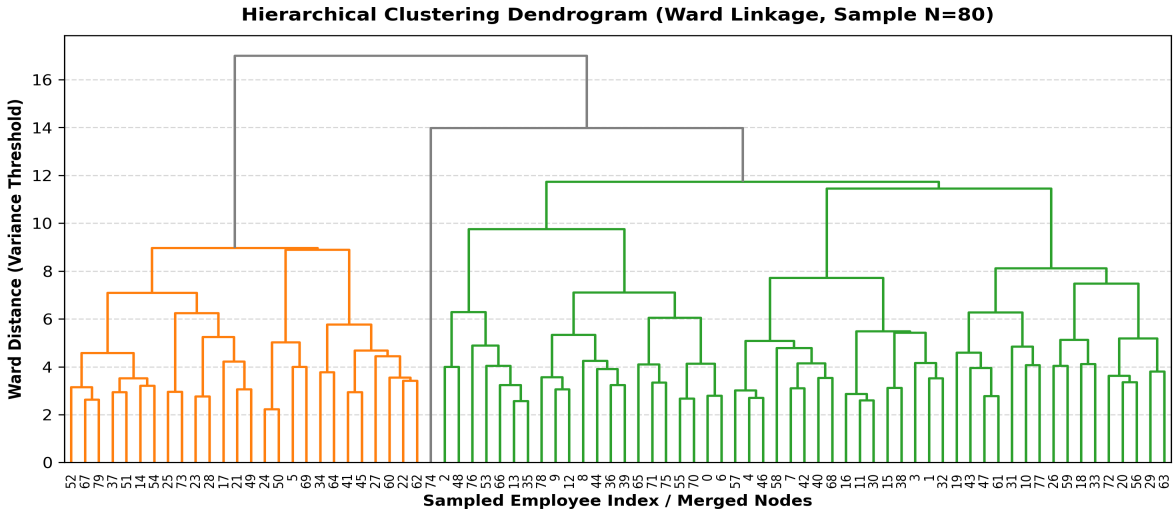
6. K-Means Clustering Methodology & Evaluation

K-Means was evaluated across k=2 through k=6 using Inertia, Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. While k=2 mathematically maximizes silhouette separation, k=4 was selected as the optimal operational configuration because it balances statistical compactness with meaningful business granularity.



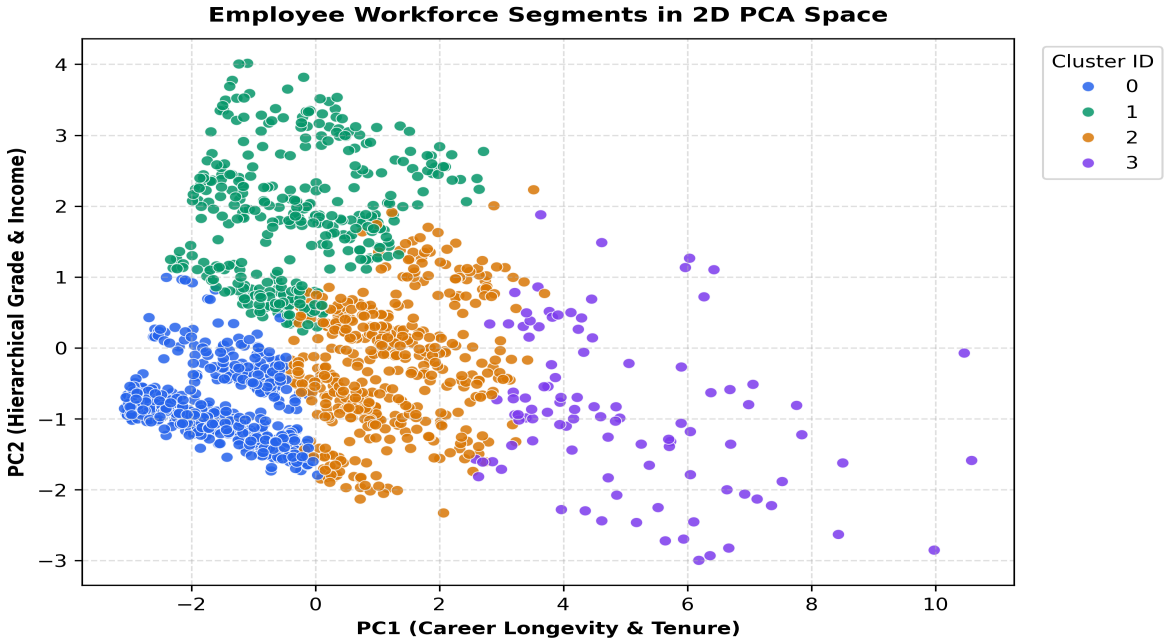
7. Hierarchical Clustering (Ward Linkage)

Agglomerative Hierarchical Clustering was independently executed using Ward minimum variance linkage and Euclidean distance. A SciPy dendrogram reveals progressive bottom-up tree merges. Cross-comparison with K-Means demonstrates substantial structural alignment (Adjusted Rand Index = 0.288, Normalized Mutual Information = 0.395).



8. Principal Component Analysis (PCA) & 2D Visualization

Principal Component Analysis decomposed the 35 transformed features into orthogonal components. PC1 (26.5% variance) represents Career Longevity and Tenure, while PC2 (11.1% variance) represents Hierarchical Grade and Compensation. Combined, the first two components capture 37.6% of total workforce variance.



9. Workforce Segment Profiles

Cluster ID	Segment Name	Headcount	Workforce Share Pct	Avg Monthly Income	Avg YearsAt Company
0	Early-Career High-Engagement Talent	557	37.89	3941.79	1.69
1	Mid-Level Professional Contributors	358	24.35	14238.65	2.33
2	Experienced Core Operational Backbone	455	30.95	9030.57	5.36
3	Senior Leadership & Tenured Specialists	100	6.80	12250.19	12.84

10. Department-Level Workforce Segmentation

Department	Headcount	Share Pct	Avg Tenure Years	Avg Monthly Income
Human Resources	107	7.28	3.36	8664.96
Research & Development	952	64.76	3.76	8287.92
Sales	411	27.96	3.80	9269.32

11. Retention & Engagement Risk Diagnostics

A composite indicator-based diagnostic was formulated from job-related factors: low satisfaction (<=2), low work-life balance (<=2), overtime intensity, and promotion stagnation (>=4 years). Observed historical separation rates are reported solely as descriptive organizational baselines; clustering models do not predict individual departure likelihood.

Cluster ID	Segment Name	Headcount	Composite Review Index	Review Priority
3	Senior Leadership & Tenured Specialists	100	38.85	Elevated Review Recommended
2	Experienced Core Operational Backbone	455	31.69	Moderate Attention
1	Mid-Level Professional Contributors	358	26.75	Moderate Attention
0	Early-Career High-Engagement Talent	557	23.90	Stable Engagement

12. Strategic HR & Business Recommendations

1. Senior Leadership & Specialists: Formalize executive knowledge transfer, establish emeritus consulting agreements, and implement retention grants for critical intellectual property.
2. Early-Career Talent: Accelerate competency growth through structured 18-month rotational programs, cross-departmental hackathons, and dedicated executive mentorship.
3. Stagnant / Retention Review Segment: Audit overtime workloads immediately, establish transparent promotional rubrics, and provide lateral career enrichment pathways to eliminate developmental bottlenecks.
4. Core Operational Backbone: Maintain competitive market compensation adjustments, recognize consistent contributors, and provide hybrid work flexibility to sustain long-term engagement.

13. Ethical Considerations, Responsible AI & Limitations

Responsible Use Mandate: *Cluster assignments reflect aggregate statistical patterns, not individual character or competence. Cluster labels must NEVER be employed as automated criteria for termination, demotion, denial of promotion, or disciplinary action. Unsupervised models identify correlations, not causation. All insights must be paired with human HR qualitative appraisal.*

14. Conclusion & Key References

Unsupervised machine learning successfully segmented the workforce into 4 distinct, interpretable talent archetypes. Dimensionality reduction via PCA confirmed that career tenure and hierarchical compensation are the predominant axes of employee differentiation. By coupling clustering outputs with supportive, non-punitive talent programs, enterprise HR can proactively foster workforce engagement and organizational resilience.

References:

• MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations.
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• Jolliffe, I. T. (2002). Principal Component Analysis. Springer Series in Statistics.
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